Rohail Ahmad

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RESEARCH INTERESTS

Primary: Trustworthy AI, Mechanistic Interpretability, Representation Learning, and Empirical AI Alignment.
Secondary: Speech Synthesis & Audio Processing for Low-Resource Languages, Deep Learning Theory.

EDUCATION

Pak-Austria Fachhochschule Institute of Applied Sciences and Technology (PAF-IAST) Haripur, Pakistan
Bachelor of Science in Artificial Intelligence 2022 – 2026
· **Undergraduate Thesis:** *Urdu Text-to-Speech System with Voice Cloning* (Supervisors: Dr. Arshad Iqbal, Dr. Muhammad Murtaza)
· **Academic Level:** EQF Level 6

OeAD Austria's Agency for Education and Internationalisation Vienna, Austria
Ernst Mach Scholarship Recipient (Austrian Government Scholarship) Sep 2025 – Feb 2026
· Awarded competitive national scholarship in Computer Science for international academic exchange in Austria.

RESEARCH & ACADEMIC PROJECTS

Urdu Text-to-Speech Synthesis & Voice Cloning Pipeline 2025 – 2026
Researcher / Undergraduate Thesis

- Engineered an end-to-end speech synthesis pipeline by fine-tuning ChatterBox TTS v0.1.2 on a curated 20,000-sample Urdu audio dataset (∼30+ hours, 24kHz WAV), achieving a convergence loss of 0.0637 and 3.04s sentence latency.
- Integrated a zero-shot Voice Encoder module for speaker adaptation and voice cloning from short reference audio.
- Built a full-stack platform (FastAPI, React, SQLite) to run user acceptance testing (10/10 rating) and qualitative MOS evaluation.

Neural Network Verification & Representation Analysis 2026 – Present
Independent Research / Ongoing IST Austria Prep

- Reimplementing local interaction neural network (LGN) architectures and transformer baselines in PyTorch to analyze activation spaces, residual stream representations, and internal feature geometry.
- Investigating geometric properties and Lipschitz bounds of neural representations to evaluate model robustness and suitability for formal input verification.
- Maintaining an open-source technical research environment focused on research-to-code execution, mechanistic interpretability, and model probe evaluations.

RESEARCH & INDUSTRY EXPERIENCE

IF/ELSE Intelligence Islamabad, Pakistan
Co-Founder & Chief Operating Officer (COO) Jul 2026 – Present

- Co-founded a B2B AI solution company delivering custom ML pipelines, generative AI agents, NLP systems, and computer vision integration for international startups and SMEs.
- Oversee technical architecture execution and lead business development with plans to establish IF/ELSE Labs, a future R&D wing aimed at transforming validated client bottlenecks into scalable, proprietary AI products.

Sybrid Pvt Ltd Islamabad, Pakistan
Data Science Intern - Innovation Department Jun 2025 – Aug 2025

- Evaluated neural TTS architectures (SpeechT5, XTTS-V2) on acoustic quality, zero-shot adaptation, and inference reliability for cross-lingual synthesis.
- Fine-tuned SpeechT5 models on curated audio data, analyzing how transcription noise and sample-rate consistency impact downstream performance.

Advance Telecom Services (ATS) Washington, United States (Remote)
Data Science Intern Oct 2023 – Nov 2023

- Performed data cleaning, preprocessing, and exploratory data analysis on Electric Vehicle (EV) datasets, training predictive machine learning models to analyze adoption and usage trends.

- Designed interactive analytics dashboards in PowerBI and leveraged Microsoft Azure cloud tools for data processing and workflow integration.

AWARDS & HONORS

- **Ernst Mach Grant:** Federal Ministry of Education, Science and Research (BMBWF), Austria (2025).
- **Award of Excellence:** Top Student of Batch 2022, PAF-IAST.
- **Certificate of Excellence:** General Secretary, AICE Conference Organizing Committee (2024).
- **Shield Award:** Deputy Campus Director, Hult Prize (2024).

ACADEMIC SERVICE & LEADERSHIP

Critical Literacy Circle of Artificial Intelligence (CLCAI) PAF-IAST
Founder 2022 – Present

– Founded and led an academic reading circle dedicated to evaluating technical AI safety papers, alignment threat models, and governance frameworks.

AI Society, PAF-IAST PAF-IAST
General Secretary Dec 2023 – Nov 2024

– Coordinated technical workshops, research seminars, and community events for over 300 students.

Hult Prize, PAF-IAST PAF-IAST
Deputy Campus Director Dec 2023 – Feb 2024

– Managed cross-functional teams, delegated operational responsibilities, and led campus-wide competition execution.

TECHNICAL SKILLS

– **Languages:** Python, C++, JavaScript, SQL, LaTeX.
– **Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, FastAPI, OpenCV, Matplotlib.
– **Domains:** Trustworthy AI, Speech Synthesis (TTS), Mechanistic Interpretability, Empirical Model Evaluation, Signal Processing.
– **Tools & Infrastructure:** Git/GitHub, Docker, Linux, Azure, Google Colab, React.

LANGUAGES

Pashto: Native · **Urdu:** Advanced (C1) · **English:** Professional (B2/C1) · **German:** Basic (A1)

REFERENCES

– **Dr. Arshad Iqbal** – Assistant Professor, PAF-IAST · Arshad.iqbal@spcai.paf-iast.edu.pk
– **Ihtisham Ali** – Assistant Professor, PAF-IAST · ihtisham.ali@spcai.paf-iast.edu.pk